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Summer Term 2026

Pattern Matching and Machine Learning for Audio Signal Processing
Exercise sheet 3

To be uploaded in eCampus till: 09-05-2026 22:00 (strict deadline)

Exercise 3.1 [$4+4+3=11$ points]

Let $N=512$, $t \in \{k/N | k \in [0:N-1]\}$ and

$$f(t) := \cos(4\pi t) + 4\cos(10\pi t) + 8\cos(2\pi 20 t).$$

(a) Explicitly write down $\boldsymbol{f} = (\boldsymbol{f}_0, \cdots, \boldsymbol{f}_{N-1}) \in \mathbb{C}^N$ in terms of the complex exponential functions $\boldsymbol{u}_k$ defined in the lecture.

(b) Calculate by hand $|\widehat{\boldsymbol{f}}(k)|$.

(c) Write a Matlab / Python function which plots $f(t)$ and its magnitude spectrum. As stated on exercise sheet 1, you should hand in a jupyter notebook file or a .m file.

Exercise 3.2 [4 points]

Calculate by hand the spectrum X of the vector $x \in \mathbb{C}^4$ defined by $x := [2, 1, -1, -2]^T$.

Exercise 3.3 [5 points]

Write a Matlab / Python function that plots the N^{th}-roots of unity and the primitive N^{th}-roots of unity for a given N. Test your function for $N = 6$.